

Sourabh Joshi

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Professional Summary

Ph.D. scholar in Electrical Engineering (Power Electronics) at IIT Roorkee with an M.Tech in Power Electronics and Drives from VNIT Nagpur. Research interests include power electronic converters, wide-bandgap semiconductor devices (SiC/GaN), battery management systems, and AI-driven energy systems. Proficient in MATLAB/Simulink, Python, and machine learning, with research experience in Li-ion battery state estimation, an IEEE conference publication, and prior industrial experience in electrical maintenance and automation.

Education

Indian Institute of Technology, Roorkee <i>Ph.D. in Electrical Engineering (Power Electronics)</i>	July 2026 – Expected 2030 CGPA: 9/10
Visvesvaraya National Institute of Technology (VNIT Nagpur) <i>M.Tech in Power Electronics & Electrical Drives</i>	July 2024 – June 2026 CGPA: 7.20/10
Maulana Azad National Institute of Technology (MANIT Bhopal) <i>B.Tech in Electrical Engineering</i>	July 2017 – May 2021 CGPA: 6.91/10

Experience

Indian Institute of Technology Roorkee (IIT Roorkee) <i>Junior Research Fellow (JRF) – ISRO Sponsored Project</i>	Jul 2026 – Present Roorkee, Uttarakhand
- Working on the ISRO, Bengaluru sponsored research project titled "<i>Enhanced Protection Enabled GaN-Based High Voltage Electronic Power Conditioner for Travelling Wave Tube Amplifiers (TWTAs) for Aerospace Applications</i>" - Conducting research on GaN-based high-voltage power electronic converters and electronic power conditioner (EPC) architectures for aerospace applications. - Designing and analysing high-frequency isolated DC–DC converter topologies, protection techniques, control strategies, and high-reliability power conversion systems for space applications. - Contributing to the development of efficient, compact, and fault-tolerant power conditioning systems for Travelling Wave Tube Amplifiers used in satellite communication payloads.	
MSP Steel and Power Limited <i>Graduate Engineering Trainee, Pellet Division – Electrical Department</i>	Jan 2022 – Jan 2023 Raigarh, Chhattisgarh
- Supported operation, preventive maintenance, and troubleshooting of plant electrical systems in a pellet production environment. - Worked with MCCs, transformers, motors, starters, automation systems, and electrical distribution equipment used in continuous industrial operations. - Assisted fault identification, corrective maintenance coordination, and equipment reliability activities for production-critical electrical assets.	

Projects

Deep Learning Based SoC and SoH Estimation for Li-Ion Battery BMS	July 2025 – April 2026
- Developed AI-based battery state estimation models for Li-ion BMS applications, targeting SoC and SoH prediction under varying operating and aging conditions. - Implemented and compared LSTM, GRU, and hybrid CNN–LSTM architectures using voltage, current, temperature, and degradation-related battery data. - Designed an Enhanced Genetic Algorithm for automated hyperparameter optimization to improve model accuracy, convergence stability, and robustness. - Evaluated models using RMSE, MAE, and R^2 metrics and benchmarked performance against conventional battery state estimation approaches.	
Evaluation of Wide-Bandgap Devices for EV Power Electronics	May 2026 – June 2026
- Evaluated Si, SiC, and GaN power semiconductor devices for EV traction inverters, onboard chargers, DC fast chargers, and auxiliary DC–DC converters. - Analyzed conduction losses, switching losses, thermal behavior, packaging constraints, EMI concerns, and reliability challenges of wide-bandgap devices. - Prepared application-oriented comparison frameworks linking device selection with efficiency, power density, thermal management, and system-level performance.	
BLDC Motor Speed Control Using MATLAB/Simulink	Jan 2020 – May 2020
- Built a closed-loop speed control model for a BLDC motor using a PID controller in MATLAB/Simulink. - Tuned controller parameters to improve transient response, reduce steady-state error, and shorten settling time in simulation.	

Publication

A Comparative Analysis of Various Deep Learning Models for State-of-Charge Estimation in Lithium-Ion Batteries

- Presented at IEEE STPEC 2025, NIT Goa, Dec 10–13, 2025.

Certifications

Supervised Machine Learning: Regression and Classification , DeepLearning.AI Coursera Certificate	May 2025
Introduction to Power Electronics , University of Colorado Boulder Coursera Certificate	Mar 2025
Arena FIDE Master , International Chess Federation (FIDE)	Dec 2021

Technical Skills

Programming and Control: Python, MATLAB, C, DSP programming, real-time control implementation

Simulation and Modeling: PLECS, MATLAB/Simulink, circuit-level simulation, converter modeling, control-system modeling, machine learning, deep learning, data preprocessing, model validation

Power Electronics: DC–DC converters, isolated DC–DC converters, high-frequency converters, resonant converters, inverters, PWM techniques, soft switching, high-frequency switching, power converter loss analysis

Magnetics and Converter Design: High-frequency transformer design, planar transformer concepts, inductor design, magnetic component selection, core selection, winding design, flux-density analysis, core and copper loss estimation

Wide-Bandgap Power Devices: GaN HEMTs, SiC MOSFETs, wide-bandgap semiconductor devices, high-frequency switching, gate-drive requirements, switching and conduction loss analysis

Digital Control and Embedded Systems: TI C2000 microcontrollers, TMS320F28379D, DSP programming, ePWM generation, ADC interfacing, dead-time generation, real-time control, digital control implementation

Hardware Design and Prototyping: PCB design, schematic development, PCB layout, component selection, SMD components, through-hole components, soldering, hardware debugging, oscilloscope-based measurements, power-stage prototyping

Battery and Energy Systems: Li-ion batteries, battery management systems, battery energy storage systems, SoC estimation, SoH estimation, EV charging systems

Power Electronics and Drives: Electrical drives, motor control, BLDC motors, inverter-fed drives, control systems, converter-fed motor drives

Industrial Electrical Systems: MCCs, transformers, motors, automation systems, electrical troubleshooting, preventive maintenance

Achievements, Memberships, and Languages

- GATE 2026: All India Rank 4032
- GATE 2024: All India Rank 5151
- JEE Main 2017: All India Rank 14780
- IEEE Graduate Student Member and IEEE Power Electronics Society (PELS) Member
- Languages: English, Hindi, Dutch